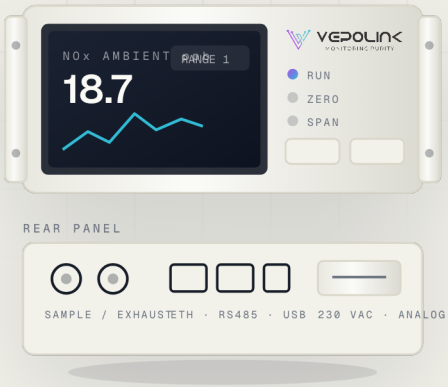
VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS ANALYSIS

CONTINUOUS AMBIENT AIR QUALITY MONITORING

Ambient NOx Gas Analyzer *Chemiluminescence*

Nitric oxide, nitrogen dioxide and total NOx measured by chemiluminescence — with a high-efficiency converter, automatic zero correction and a year of logged data on board.



0.1 · 0.5 · 1 ppm
USER-SELECTABLE RANGES

7 inch
TFT LCD TOUCHSCREEN

1 year
ON-BOARD DATA LOGGING

CHEMILUMINESCENCE REFERENCE METHOD | **NO_2-NO CONVERTER** HIGH EFFICIENCY | **CPCB** ALIGNED REPORTING

0.5 ppb
LOWER DETECTABLE LIMIT

3
SELECTABLE MEASURING RANGES

60 s
TYPICAL RESPONSE TIME T90

32 bit
ARM PROCESSOR, LINUX BASED

01 OVERVIEW

Nitrogen oxides, *resolved to NO and NO_2.*

A continuous ambient NOx analyzer built on the chemiluminescence method — the accepted technique for regulatory NO, NO_2 and total NOx measurement. An internal ozone generator sized for long, stable service and a high-efficiency NO_2-to-NO converter cartridge give a low-drift reading across all three reported channels.

The instrument runs a 32-bit ARM processor on a Linux kernel with a 7-inch touchscreen interface, an automatic zero correction system for unattended stability, and Ethernet, RS-232, RS-485, USB and analog interfaces so the analyzer reports into a station, a SCADA system and a regulatory server at the same time.

REPORTED CHANNELS

- NO INSTANT
- NO_2 INSTANT
- NOX TOTAL
- 1-MIN AVERAGE
- HOURLY AVERAGE
- 24-HR AVERAGE
- SAMPLE FLOW
- CONVERTER EFFICIENCY
- ZERO & SPAN STATUS
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Durable Ozone Generator

An internal O_3 generator designed for strong durability keeps the chemiluminescence reaction consistent for a long service life without failure.

High-Efficiency NO_2 Converter

The NO_2-to-NO converter uses a high-efficiency cartridge material with a conversion rate approaching 100%, so derived NO_2 stays accurate.

Touchscreen Operation

A 7-inch TFT LCD touch interface puts live NO, NO_2 and NOx values, trends, ranges and diagnostics in front of the operator.

One Year of Data On Board

An internal NAND-flash logger holds up to a year of one-minute averages, downloadable over USB or network.

02 MEASUREMENT PRINCIPLE

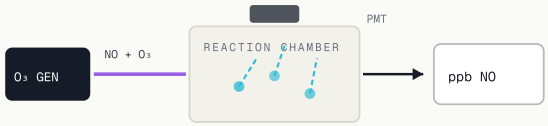
How chemiluminescence *counts NOx.*

Nitric oxide reacts with ozone to form excited nitrogen dioxide, which relaxes by emitting light in proportion to concentration — a direct optical measurement with no reagents in the sample path beyond the ozone generator itself.

CHEMILUMINESCENCE

React, Emit, Count

O_3 GENERATOR · REACTION CHAMBER · PHOTOMULTIPLIER



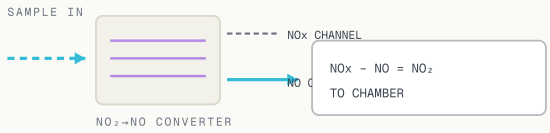
Sample NO reacts with ozone generated inside the analyzer. The excited NO_2 formed relaxes by emitting near-infrared light; a photomultiplier counts that emission and the analyzer converts intensity to concentration.

`Signal ∝ NO concentration`

NO_2 CONVERSION

Converter Channel

HIGH-EFFICIENCY NO_2→NO CONVERTER



A second sample path runs through a converter cartridge that reduces NO_2 to NO ahead of the chamber, giving total NOx. Subtracting the direct NO channel from the NOx channel yields NO_2.

`NOx − NO = NO₂`

STABILITY

Auto Zero & Compensation

AUTOMATIC BASELINE · T AND P CORRECTION



An automatic zero correction cycle re-establishes the baseline without operator intervention, while temperature and pressure compensation and a durable ozone generator hold the reading steady through the day.

`Unattended long-term operation`

DATA PATH

Logging & Reporting

ETHERNET · RS-232 / 485 · USB · ANALOG



Readings are stored internally as one-minute averages for up to a year and transmitted live to a station data logger, plant SCADA or a pollution control board server.

`Download over USB or network`

03 SAMPLE PATH

From inlet to *reading.*

Everything between the sample line and the display is designed to separate NO from NO_2 cleanly and keep the ozone reaction consistent, run after run.

STEP 01

Filtered intake

Ambient air is drawn through a particulate filter and inert sample line into the analyzer at a controlled flow.

STEP 02

Dual-channel split

The sample splits into a direct NO channel and a NOx channel routed through the NO_2-to-NO converter.

STEP 03

Ozone reaction

Both streams react with internally generated ozone in the chamber; excited NO_2 emits light proportional to concentration.

STEP 04

Detect and report

Chemiluminescence is counted on both channels, NO_2 derived by subtraction, corrected, displayed, logged and transmitted.

WHY CONVERTER EFFICIENCY MATTERS

The NO_2-to-NO converter's efficiency sets the accuracy of every derived NO_2 and total NOx value. The cartridge material is chosen for a conversion rate approaching 100%, so drift in that single component does not creep into every downstream reading.

RANGE SELECTION

Three ranges — 0–0.1, 0–0.5 and 0–1 ppm — are user selectable from the touchscreen. Urban background sites usually run on the lowest range for resolution; traffic corridors and near-source installations use a wider range so episodes are captured without saturating.

WHY NITROGEN OXIDES ARE MONITORED

NOx forms wherever fuel is burned at high temperature — vehicle engines, power generation and industrial furnaces are the dominant sources. NO_2 irritates the airways directly, and together with volatile organic compounds and sunlight it drives the formation of ground-level ozone and photochemical smog. Continuous, speciated measurement of NO and NO_2 is what lets a city or a plant separate combustion contribution from background and act on it.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

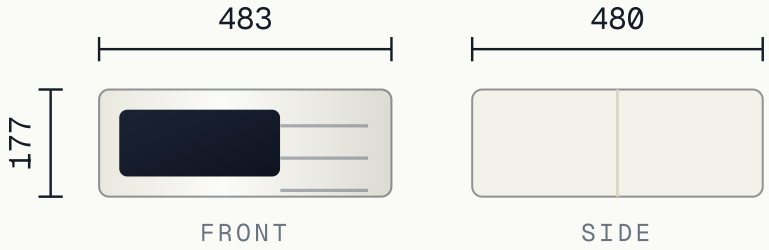
Parameter	Ambient NO, NO_2 and total NOx
Working Principle	Chemiluminescence
Measuring Ranges	0 - 0.1 / 0.5 / 1 ppm, user selectable
Lower Detectable Limit	0.5 ppb
Precision	±1 % of reading
Linearity	±1 % of full scale
Zero Drift	< ±1 ppb per 24 hours
Span Drift	< ±1 % FS per 7 days
Response Time	T90 typically 60 s
NO_2–NO Converter	High-efficiency cartridge, near 100 % conversion
Ozone Generator	Internal, long-life, low-maintenance

SYSTEM · COMMS · BUILD

Display	7-inch TFT LCD with touch interface
Controller	32-bit ARM Cortex-M4, Linux based multiprocessing
Zero Correction	Automatic · internal zero optional
Compensation	Temperature and pressure
Data Logger	NAND flash · up to 1 year of 1-minute averages
Interfaces	Ethernet · RS-232 · RS-485 · USB · analog out
Data Download	USB export or network transfer
Power Supply	230 VAC · 50 Hz
Mounting	19-inch rack or bench, station cabinet
Operating Conditions	5 - 40 °C, non-condensing
Enclosure	Weather-protected station cabinet for outdoor sites

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



05 INSTALLATION & UPKEEP

Racked, calibrated, *left running.*

The analyzer is installed in a station cabinet with the sample manifold, calibration source and data logger. Per CPCB practice, the sample inlet sits in free-flowing ambient air **3 to 10 m above ground**, with an inert, short sample line and a scheduled **zero, span and converter-efficiency check** traceable to certified gas.

01 · MOUNTING

19-inch rack or bench

Standard rack fitment alongside other CAAQMS analyzers, sharing the station's manifold, power and data logger.

02 · SAMPLE LINE

Inert and short

PTFE line with a particulate filter, kept short so NO and NO_2 reach the chamber and converter without loss.

03 · ZERO AIR

Clean baseline source

A zero air unit supplies scrubbed air for the automatic zero cycle and for periodic manual verification.

04 · CALIBRATION

Multipoint traceable

Span with certified NO and NOx gas from a cylinder via a multipoint calibrator, on the site's audit schedule.

05 · CONSUMABLES

Filter and converter

Routine upkeep is the inlet filter, periodic leak check and converter cartridge and ozone generator inspection. No wet chemistry.

06 · DIAGNOSTICS

Reported with the data

Converter efficiency, chamber temperature and pressure, flow and zero/span status travel alongside every reading.

PART OF A FULL STATION

The NOx analyzer is normally specified with SO_2, CO and ozone analyzers, a multipoint calibrator and zero air unit, a particulate monitor and a meteorological mast. Sharing a time base, cabinet and data path, they report as one station record — which is what a pollution control board expects to receive.

COMMISSIONING & AUDIT

Commissioning covers leak and flow verification, a multipoint linearity check across the working range, a converter-efficiency check against a known NO_2 source, and a documented zero and span baseline that becomes the reference for subsequent audits. Thereafter the automatic zero cycle carries day-to-day stability and the scheduled span check confirms it against certified gas, so the record stays defensible without a technician on site.

06 APPLICATIONS

Where it *works.*

Wherever combustion runs hot, NOx has to be measured — for compliance, for traffic and urban air quality reporting, and for understanding what a source actually contributes downwind.

Regulatory CAAQMS Networks	Thermal Power Stations	Urban & Traffic Corridors
Refineries & Petrochemicals	Chemical & Fertiliser Plants	Automotive & Vehicle Testing
Ports & Marine Fuel Zones	EIA & Baseline Studies	Research & Air Quality Studies

WHAT THE DATA SUPPORTS

Continuous, speciated NOx records serve three audiences at once: regulators, who need NO, NO_2 and NOx against national standards; operations, who need to see the effect of combustion tuning, fuel switching and load changes; and communities near a corridor or fence line, who need a credible answer about the air they breathe. One measurement, three conversations.

SITING NEAR A SOURCE

Near-road and near-source installations are usually placed at breathing height close to the emission point, with a background station upwind or away from traffic so the source's contribution can be separated from what arrives already in the air. Where load or traffic volume varies sharply, a wider measuring range keeps peaks on scale rather than clipped.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Chemiluminescence. Nitric oxide reacts with internally generated ozone to form excited NO_2, which emits light in proportion to concentration, detected by a photomultiplier and reported as NO, NO_2 and NOx.

Q Which parameters does it measure?

NO, NO_2 and total NOx concentrations, for continuous ambient air quality monitoring in regulatory, industrial and research applications.

Q Is it suitable for continuous monitoring?

Yes. It is designed for real-time, uninterrupted operation, with automatic zero correction so the baseline holds without an operator present.

Q Does it meet CPCB requirements?

The analyzer is designed to meet CPCB and regulatory ambient air quality monitoring guidelines, and reports in the formats a state or central board server expects.

Q Can it be installed outdoors?

Yes, housed in a weather-protected station cabinet. The analyzer itself is a rack instrument; the cabinet provides the environmental protection and shared services.

Q How is NO_2 measured if it isn't detected directly?

A converter reduces NO_2 to NO ahead of a second reaction, giving a total NOx reading; subtracting the direct NO channel from that total yields NO_2.

Q What ranges are available?

0–0.1, 0–0.5 and 0–1 ppm, selectable by the user from the touchscreen to suit background, urban or near-source conditions.

Q How much data does it hold?

The internal NAND-flash logger stores up to one year of one-minute averaged data, exportable over USB or pulled across the network.

Q How does it integrate with a station?

Ethernet, RS-232, RS-485 and analog outputs let the analyzer report to a station data logger, plant SCADA and a regulatory server simultaneously.

Q What maintenance does it need?

Inlet filter changes, periodic leak and flow checks, converter cartridge and ozone generator inspection, and scheduled zero and span verification with certified gas.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

NO, NO_2 AND NOX, MEASURED CONTINUOUSLY

Know your NOx. *In real time.*

Tell us the site, the range you expect and who receives the data, and we'll configure the analyzer and the station around it — calibration, zero air, cabinet and reporting. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, installs and calibrates ambient gas analyzers. No call centre.
+91 99538 86178 →

02 **Request a station configuration**
RECOMMENDED
Send your site and reporting requirement — we'll spec ranges, calibration, zero air, cabinet and communications.
sales@vepolink.com →

03 **Book a demonstration**
See a running analyzer and the reporting dashboard with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM